

Zero-Shot Large Audio-Language Models versus Trained Captioners: A Preregistered Evaluation on Clotho with Polyphony and Hallucination Analyses

Zuraiz

M.Sc. International Software Systems Science, University of Bamberg

CH-Proj-M: Master's Project Computational Humanities, SS 2026

Supervisor: Prof. Dr.-Ing. Jakob Abeßer

Abstract—Large audio-language models (LALMs) promise open-vocabulary audio understanding without task-specific training, but whether they match trained audio captioners remains unclear. We evaluate three zero-shot LALMs (Qwen2.5-Omni-7B, SALMONN-13B, Audio Flamingo 3) against three traditional systems (an AST tagging floor, the DCASE 2023 CNN14+BART baseline, EnCLAP-base) on the full Clotho v2.1 evaluation split under one harness with identical data, decoding provenance, and DCASE metrics. The CNN14 baseline reproduces its published score within 0.002 SPIDeR-FL, validating the harness; the EnCLAP row runs ≈ 0.011 below its published value, a gap we report and examine. Zero-shot performance is strongly model-dependent: two LALMs trail both trained captioners, while Audio Flamingo 3 surpasses them on every metric under the same harness (SPIDeR-FL 0.297 vs. 0.280, also above EnCLAP's published 0.291; bootstrap CI excludes the preregistered 0.261 threshold). A PANNs-based polyphony split shows *higher* caption-overlap scores on event-rich polyphonic clips for all six systems, and a closed-vocabulary hallucination metric (CHAIR-audio) shows the best captioner does not hallucinate less per caption than its LALM peer, although per mention it is the most grounded system in the table. “Zero-shot” denotes the evaluation protocol (no fine-tuning, no in-context examples); the audio-specialist LALMs’ training corpora include Clotho development pairs, a caveat the paper examines. All hypotheses were preregistered; deviations are disclosed. Code: <https://github.com/Zuraiz270/audio-to-text-captioning-using-LALMs>.

I. INTRODUCTION

Automated audio captioning (AAC) maps an audio recording to a natural-language description of its sound events. Traditional AAC systems are trained end-to-end on caption corpora such as Clotho [1]; large audio-language models (LALMs) instead couple a pretrained audio encoder to a large language model and are used *zero-shot*.

The task is hard for two reasons. Real recordings rarely contain one clean source: everyday scenes stack engines, voices, weather, and animals on top of each other, and a captioner must both detect this mixture and verbalise it. Evaluation is just as delicate, because reference-overlap metrics reward phrasing that resembles five particular human sentences, which is not the same thing as being acoustically right. Reliable captioning would nevertheless be useful well

beyond benchmarks, for indexing large sound archives, for urban and environmental monitoring, and for making audio material accessible in text form. Whether zero-shot LALMs already match purpose-trained captioners, and where they fail, is the question this project answers empirically. We ask: **RQ1**: Do zero-shot LALMs match or exceed traditional captioning baselines on standard metrics (SPIDeR-FL, CIDEr)? **RQ2**: How does performance differ between polyphonic (overlapping events) and monophonic clips? **RQ3**: What are the failure modes, in particular caption hallucination?

Our contributions are: (i) a reproducible six-system comparison on the full Clotho-eval split, anchored by a CNN14 reproduction that validates the harness; (ii) a preregistered, bootstrap-based hypothesis-testing protocol with all deviations disclosed; (iii) an SED-derived polyphony split and a closed-vocabulary adaptation of the CHAIR hallucination metric [2] to audio. The paper reviews related work in Sec. II, describes the harness, the polyphony split, the hallucination metric, and the statistics in Sec. III, reports results per research question in Sec. IV, and closes with discussion, limitations, and conclusions in Secs. V and VI.

II. RELATED WORK

Datasets, systems, and metrics. Clotho v2.1 [1] is the standard AAC benchmark: 1,045 evaluation clips (15–30 s, 44.1 kHz) with five human captions each. The DCASE challenge series established SPIDeR [3] and its fluency-penalised variant SPIDeR-FL [4] as headline metrics; SPIDeR-FL’s fluency detector also underlies FENSE, the caption metric used in [5]. We compute all metrics with `aac-metrics` [6]. Trained captioners include the DCASE 2023 baseline (CNN14+BART, encoder from PANNs [7]) [8] and EnCLAP [9], whose released checkpoints were later re-measured by its authors [10]. AST [11] is a 527-class AudioSet [12] tagger we use as a tagging floor.

LALMs. Qwen2.5-Omni [13] is a general text/vision/audio/speech model; SALMONN [14] fuses Whisper and BEATs encoders through a Q-Former into a 13B LLM; Audio Flamingo 3 (AF3) [15] pairs a Whisper-style

encoder with a Qwen2-7B decoder and is the current open state of the art for audio understanding. Together they cover a general omni-model, an earlier audio specialist, and a current audio specialist from three different vendors.

Polyphony and hallucination. For polyphonic scene understanding, Abeßer et al. [16] introduced the USM dataset, and Harish and Abeßer [5] proposed a three-level protocol (tagging, counting and loudness ranking, captioning) on which LALM performance on *event-level* tasks degrades with increasing polyphony, a finding our *caption-level* RQ2 complements. CHAIR [2] measures object hallucination in image captioning against a closed list; MACE [17] scores audio captions against the audio itself via CLAP embeddings.

Our study differs from these efforts in three ways. It evaluates on real Clotho recordings rather than synthetic mixtures, so polyphony comes from the world and not from a mixing script. It anchors the LALMs against trained captioners reproduced inside the same harness, which turns cross-paper number comparisons into one controlled measurement. And it commits to hypotheses, thresholds, and fallback rules in advance and reports every deviation, which is still uncommon in audio captioning evaluations.

III. METHOD

A. Evaluation Harness

Every system implements one contract, `caption(waveform, sr) → str`, behind a registry, so a single inference loop and a single scorer serve all rows. Each run writes a predictions JSON plus a manifest recording checkpoint SHA-256s, library versions, decode parameters, and the seed (42). Scoring uses the `dcase2023` preset of `aac-metrics` 0.5.5 (SPIDER-FL, CIDEr-D, SPICE, METEOR, fluency error) on identical clip sets; a `-subset` mechanism restricts scoring to fixed clip lists for same-set comparisons. Baselines ran on CPU; LALMs ran on one NVIDIA A100-40GB each (offline, pre-cached checkpoints).

Runs are engineered for auditability. The inference loop flushes predictions atomically every ten clips and isolates per-clip failures, so a crash loses at most nine captions and one broken file cannot abort a run. SPICE requires Java 8 to 13, so scoring lives in a pinned Linux environment while inference runs natively; the predictions JSON is the only handoff between the two, and between laptop and cluster. Cluster compute nodes have no internet access, which forces fully offline model loads from pre-cached checkpoints and makes the runs repeatable by construction. Full 1,045-clip runs are cheap on one A100: 8.3 minutes for Qwen2.5-Omni (about 22 GB VRAM in bf16), 19 minutes for SALMONN (fp16), and 17.5 minutes for AF3 (about 28 GB in fp32). Each LALM receives a fixed one-line instruction asking for one short, factual sentence; the exact prompt of every row is recorded in its manifest. All 1,045 clips completed for every system with zero failures.

B. Systems

AST (tagging floor): top-5 sigmoid AudioSet labels templated as “a sound of ...”; input window 10.24 s. **CNN14+BART**: the official DCASE 2023 baseline, reproduced at 0.259 SPIDER-FL against the published 0.261; this reproduction validates the harness. **EnCLAP-base** (AudioCaps-pretrained, Clotho-finetuned release): 0.280 SPIDER-FL in our harness vs. 0.295 SPIDER published for this configuration [9] and 0.291 SPIDER-FL measured for the released base checkpoint with the same `aac-metrics` toolkit [10]. The ≈ 1.1 pp shortfall (plausibly checkpoint selection or decoding differences) is reported rather than tuned away; H1 is anchored on CNN14 and unaffected. **Qwen2.5-Omni-7B**: audio-only, talker disabled, greedy decoding, bf16. **SALMONN-13B**: Whisper + BEATs $\rightarrow$ window-level Q-Former $\rightarrow$ Vicuna-13B, fp16, beam 4 (its authored default). **AF3**: Whisper-style encoder + Qwen2-7B decoder, fp32 (its native precision), greedy. Each LALM uses its authors’ recommended decoding; all decode settings are recorded per row.

C. Polyphony Split (RQ2)

We derive pseudo-strong labels with PANNs `Cnn14_DecisionLevelMax` [7] (framewise output, empirically 100 fps). A clip is *polyphonic* iff two AudioSet classes are simultaneously active for ≥ 1 s at confidence $\geq \tau$, evaluated as the longest co-activation run of the same class pair. The project’s public definition fixed $\tau=0.50$; a fallback rule, committed *before* any subset results were seen, required ≥ 300 clips per bucket and otherwise selected the largest stored τ satisfying it. At $\tau=0.50$ only 106 clips were polyphonic (609 clips had no activation at all), so the rule selected $\tau=0.25$: **336 polyphonic / 709 monophonic**. As a validity check, the five reference captions of SED-polyphonic clips mention more distinct sound entities than monophonic ones (mean 4.5 vs. 3.8; ≥ 2 entities in 97% vs. 87% of clips).

D. CHAIR-audio (RQ3)

Vision CHAIR [2] counts caption objects absent from the image against a fixed 80-object list. We port this to audio with the 527-label AudioSet vocabulary (604 surface forms after splitting comma alternatives and dropping six generic structural forms; deterministic word-boundary matching with plural/-ing variants; no learned NLP). An entity in a model caption is *hallucinated* iff it appears neither in (a) the union of the five reference captions nor (b) the clip’s SED tag set at threshold τ (dual criterion). We report CHAIR-s (fraction of captions with ≥ 1 hallucinated entity) at $\tau \in \{0.20, 0.25, 0.30\}$ per the preregistered sensitivity bracket, with matcher coverage, and CHAIR-i (fraction of hallucinated entity mentions) as a descriptive decomposition.

E. Secondary Metric: MACE

MACE [17] scores a caption directly against the audio through CLAP embeddings and applies the FENSE fluency penalty, so it needs no reference overlap at all. We use the authors’ reference implementation with its MS-CLAP

TABLE I
FULL CLOTHO-EVAL (1,045 CLIPS). PUBLISHED SPIDER-FL ANCHORS:
CNN14 0.261 [8], ENCLAP-BASE 0.291 [10].

Model	SPIDER-FL	CIDEr-D	SPICE	METEOR	FER ↓
AST (tag template)	0.068	0.110	0.056	0.095	0.180
Qwen2.5-Omni-7B	0.188	0.286	0.091	0.141	0.005
SALMONN-13B	0.225	0.347	0.110	0.152	0.017
CNN14+BART (DCASE'23)	0.259	0.416	0.118	0.176	0.029
EnCLAP-base	0.280	0.443	0.123	0.179	0.013
Audio Flamingo 3	0.297	0.460	0.137	0.186	0.011

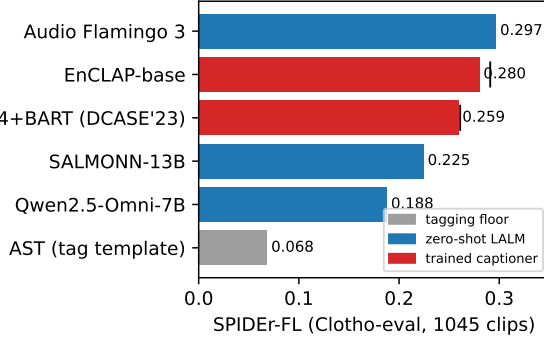


Fig. 1. SPIDER-FL on full Clotho-eval. Black ticks: published SPIDER-FL of the trained captioners (CNN14 0.261, EnCLAP-base 0.291).

backend. Two implementation properties matter for interpretation: the backend embeds a randomly cropped 7s audio window per call, which makes repeated runs vary by about 0.002, and its text encoder pads to 77 GPT-2 tokens without truncating, so we clamp candidates to 30 words, which affects 3 of 3,135 LALM captions. We treat MACE as a secondary, audio-grounded check on the reference-based results, not as a headline metric.

F. Preregistered Hypotheses and Statistics

Hypotheses were preregistered with BCa bootstrap [18] ($n=1000$ resamples, seed 42) and Holm–Bonferroni correction: **H1**: SPIDER-FL(AF3) > 0.261 (the published CNN14 baseline; minimum detectable effect 1.04 pp); **H2**: Δ SPIDER-FL(poly–mono) > 0 per LALM (MDE 1.50 pp); **H3**: AF3 > SALMONN (SPIDER-FL); **H4**: CHAIR(AF3) < CHAIR(SALMONN). The preregistration was drafted before the LALM runs but never formally frozen; we therefore treat it as a declared analysis plan and disclose every deviation (Sec. V-A) [19]. Corpus SPIDER-FL equals the mean of per-clip scores exactly, so the bootstrap tests the reported metric itself.

IV. RESULTS

A. RQ1: Zero-Shot LALMs vs. Trained Captioners

Table I and Fig. 1 give the main comparison. The ordering is $AST \ll Qwen2.5-Omni < SALMONN < CNN14 < EnCLAP < AF3$ (0.068 $\rightarrow$ 0.188 $\rightarrow$ 0.225 $\rightarrow$ 0.259 $\rightarrow$ 0.280 $\rightarrow$ 0.297 SPIDER-FL). Every LALM clears the tagging floor by a factor of 3–4, so all of them describe scenes rather than list events.

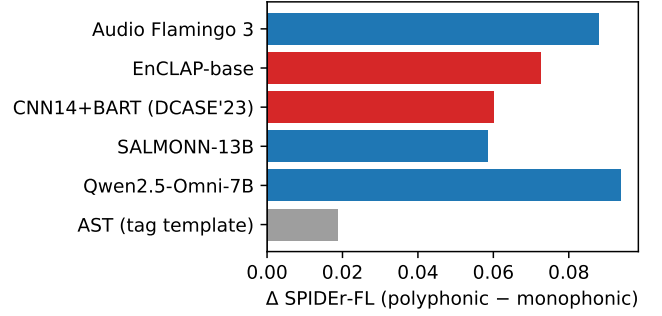


Fig. 2. Δ SPIDER-FL (polyphonic – monophonic) per system. Every bar is positive, including all three trained or tagging systems, which is what identifies the shift as a subset-difficulty effect.

Two of the three nonetheless trail *both* trained captioners; the blanket claim that zero-shot LALMs beat trained systems is false. The audio-specialist AF3, however, beats both trained captioners *on every metric* within the harness, zero-shot; its 0.297 also exceeds EnCLAP’s *published* SPIDER-FL of 0.291 [10], though not EnCLAP’s published CIDEr (0.460 vs. 0.461–0.463 [9]). **H1 is supported**: mean SPIDER-FL 0.2968, one-sided 95% BCa lower bound 0.2828 > 0.261 + MDE (0.2714), $p \approx 0.001$, significant after Holm correction ($\alpha'=0.0167$). All LALMs are highly fluent (FER ≤ 0.017 ; Qwen2.5-Omni’s 0.005 is the lowest in the table), so their differences reflect content, not disfluency. **H3 is supported**: per-clip paired AF3–SALMONN difference +0.072 (CI lower +0.058, $p \approx 0.001$).

Table II shows the register differences behind these numbers. AST emits a tag list. Qwen2.5-Omni wraps its content in report-like framing (“The audio contains ...”) and enumerates several events per clip, a style no Clotho reference uses. SALMONN and the trained captioners produce short declaratives. AF3 reads closest to the Clotho register, which is precisely what overlap metrics reward. The second clip also previews the hallucination analysis: AF3’s fluent caption invents a zipper that neither the references nor the audio tags support.

B. RQ2: Polyphonic vs. Monophonic

Table III and Fig. 2: *all six systems score higher on the polyphonic subset*. **H2 is supported for every LALM**: AF3 $\Delta=+0.088$ (CI lower +0.058), Qwen2.5-Omni +0.094 (+0.071), SALMONN +0.059 (+0.032); all $p \approx 0.001$ (AF3’s Δ , the Holm family member, passes at $\alpha'=0.05$). Because the trained baselines shift by comparable amounts (+0.060 to +0.073), this is a *subset-difficulty* effect rather than an LALM property: event-rich polyphonic clips give captions more matchable content, while the monophonic bucket concentrates quiet, ambiguous recordings (219 clips have no SED activation at $\tau=0.25$ at all) whose descriptions legitimately diverge from references. No LALM shows a polyphony *penalty* relative to the trained systems at the caption-overlap level. This complements the event-level degradation with polyphony

TABLE II

VERBATIM SYSTEM OUTPUTS FOR TWO EVALUATION CLIPS, NEXT TO ONE OF THE FIVE HUMAN REFERENCES. THE EXAMPLES PREVIEW BOTH FINDINGS: CAPTION REGISTER DIFFERS SHARPLY ACROSS SYSTEMS (RQ1), AND EVEN THE MOST FLUENT CAPTIONER INVENTS UNSUPPORTED DETAIL, HERE A ZIPPER (RQ3).

System	20070819.fjord.beach.00.wav	creaky.wav
Human reference (1 of 5)	During the storm, ocean waves were crashing and breaking against an uneven shoreline.	First, people are walking and then voices are talking in the background while music plays softly.
AST (tag template)	a sound of ocean, waves, surf, wind, wind noise and boat, water vehicle	a sound of speech, walk, footsteps, animal, inside, large room or hall and outside, rural or natural
Qwen2.5-Omni-7B	The audio contains the sound of waves crashing.	The audio contains the sounds of footsteps, a door creaking open and shut, and some indistinct voices in the background.
SALMONN-13B	Waves are crashing on a beach.	People are walking and talking in a corridor.
CNN14+BART (DCASE'23)	the rain is falling down and the wind is blowing	a person is walking on a hard surface while walking
EnCLAP-base	Waves are crashing against the shoreline of a beach.	A person is walking on a hard floor.
Audio Flamingo 3	Heavy rain falls as waves crash against the shore.	A person is walking around and talking while opening and closing a zipper.

TABLE III

SPIDER-FL ON THE POLYPHONIC VS. MONOPHONIC SUBSETS ($\tau=0.25$; 336/709 CLIPS).

Model	poly	mono	$\Delta(\text{poly}-\text{mono})$
AST (tag template)	0.081	0.062	+0.019
Qwen2.5-Omni-7B	0.250	0.157	+0.094
SALMONN-13B	0.262	0.204	+0.059
CNN14+BART (DCASE'23)	0.298	0.238	+0.060
EnCLAP-base	0.329	0.256	+0.073
Audio Flamingo 3	0.356	0.268	+0.088

TABLE IV

MACE (AUDIO-GROUNDED, MS-CLAP BACKEND) ON THE TWO SUBSETS. SAME DIRECTION AS SPIDER-FL; NOTE SALMONN AND QWEN2.5-OMNI ARE NEARLY LEVEL HERE DESPITE A 0.037 SPIDER-FL GAP.

Model	MACE poly	MACE mono	$\Delta(\text{poly}-\text{mono})$
Audio Flamingo 3	0.620	0.593	+0.028
SALMONN-13B	0.559	0.538	+0.021
Qwen2.5-Omni-7B	0.560	0.528	+0.032

reported by Harish and Abeßer [5]: counting and naming all events gets harder with polyphony, but producing a reference-like *description* does not. The audio-grounded MACE metric independently corroborates the direction on the LALMs: every poly–mono ΔMACE in Table IV is positive, so the polyphonic advantage is not an artefact of reference overlap.

C. RQ3: Hallucination

Table V: at $\tau=0.25$, AF3 (0.347), SALMONN (0.332), CNN14 (0.350) and EnCLAP (0.351) cluster tightly, while Qwen2.5-Omni hallucinates markedly more (0.550) and AST’s 0.956 is the expected validity check: an indiscriminate top-5 tag list almost always contains an unsupported label. **H4’s null is retained:** AF3 does *not* hallucinate less than SALMONN (paired difference -0.015 at $\tau=0.25$, $p=0.82$; the sign and decision are stable across all three thresholds, so the outcome is not threshold-sensitive). The best captioner by overlap

TABLE V

CHAIR-AUDIO CAPTION-LEVEL HALLUCINATION RATES (DUAL CRITERION), LOWER IS BETTER.

Model	CHAIR-s at τ			coverage ($\tau=0.25$)
	0.20	0.25	0.30	
AST (tag template)	0.952	0.956	0.968	1.00
Qwen2.5-Omni-7B	0.534	0.550	0.561	0.91
SALMONN-13B	0.322	0.332	0.337	0.80
CNN14+BART (DCASE'23)	0.343	0.350	0.358	0.75
EnCLAP-base	0.338	0.351	0.357	0.76
Audio Flamingo 3	0.331	0.347	0.355	0.82

metrics is thus not the most grounded one; higher SPIDER-FL is partly earned by richer, more specific captions that carry more hallucination surface.

Decomposing the rate explains the null. Per caption, AF3 mentions more vocabulary entities than SALMONN (1.54 vs. 1.38 on average; Qwen2.5-Omni 2.08, AST 5.49), and each extra mention is another chance to be unsupported. Per mention, the picture flips: AF3’s CHAIR-i of 0.299 at $\tau=0.25$ is the lowest in the table (SALMONN 0.314, CNN14 0.318, EnCLAP 0.320, Qwen2.5-Omni 0.437, AST 0.602). AF3 is therefore the most grounded system per claim it makes; it simply makes more claims per caption, as the zipper example in Table II illustrates. We report this as a descriptive decomposition; H4 was preregistered on the per-caption rate and its verdict stands. In constrained event tagging, by contrast, Harish and Abeßer [5] found Qwen2.5-Omni the *more precise* (less tag-hallucinating) of their two LALMs. Its high CHAIR here therefore points at unconstrained caption generation rather than at the model’s audio grounding, and its competitive MACE fits that reading. MACE, which scores captions against the audio itself, agrees with the ranking split: AF3 leads (0.593–0.620), but Qwen2.5-Omni (≈ 0.53 –0.56) draws level with SALMONN despite trailing it by 0.037 SPIDER-FL. Reference overlap and audio grounding rank models differently. A concrete failure mode: greedy decoding drove Qwen2.5-Omni into a 515-word repetition

loop (“tapped, tapped, . . .”) on one clip, caught by the fluency-error metric and visible only because per-clip outputs are retained.

V. DISCUSSION AND LIMITATIONS

What “zero-shot” does and does not mean. Zero-shot here denotes the evaluation protocol: no task-specific fine-tuning by us and no in-context examples. It does not mean Clotho-free training: AF3’s training corpus lists the Clotho development pairs (19,195 audio–text pairs) [15], and SALMONN’s pre-training and instruction tuning both include Clotho [14]; Qwen2.5-Omni’s data mix is undisclosed [13]. The evaluation split is standard held-out data for all systems, but prior exposure to the Clotho caption register plausibly contributes to AF3’s overlap-metric advantage (one reason its captions read Clotho-style). It is also the same development data the trained baselines were fitted on, so the comparison remains symmetric at the data level. Harish and Abeßer [5] sidestep this by evaluating on synthetic USM soundscapes; our audio-grounded metrics (CHAIR, MACE) serve the same role here. **Metric scope.** Overlap metrics reward reference-like phrasing; AF3’s Clotho-style captions fit them well. CHAIR-audio and MACE add orthogonal, audio-grounded axes on which AF3’s advantage shrinks or disappears. All three are needed for a fair picture. (MACE caveat: its MS-CLAP backend randomly crops a 7 s window per call, giving run-to-run variation of ≈ 0.002 .) **Polyphony split.** SED pseudo-labels inherit PANNs’ biases; AudioSet ontology parents (e.g. “Music”/“Musical instrument”) can co-fire and inflate polyphony. The reference-entity audit supports the split’s direction. The AST cross-check sees only the first 10.24 s. **CHAIR matcher.** The closed vocabulary misses synonyms outside AudioSet surface forms (coverage ≈ 0.8); this bias applies uniformly to all models and largely cancels in the paired H4 comparison. **Compute.** LALMs ran on an A100; baselines on CPU. Only the compute location differs; data, scorer, and references are identical. **Where this sits.** Harish and Abeßer’s three-level protocol places captioning at the top of an understanding hierarchy, above tagging and counting [5]. Our results fill in the caption level on real recordings, and the combined picture is consistent: at the event level polyphony hurts, at the caption level it does not, and hallucination behaves like a property of free generation rather than of audio grounding. For evaluation practice this argues for reporting one grounded metric next to every overlap score, and for stating training-data overlap whenever “zero-shot” claims are made on benchmark corpora. As an external consistency check, AF3’s own report lists a Clotho CIDEr of roughly 0.50 under its internal protocol [15], in the same range as the 0.460 our minimal one-line prompt obtains.

A. Deviations from the Preregistration

(1) The preregistration was drafted but never formally frozen (`freeze_date: null`); we disclose rather than claim confirmatory status. (2) RQ2’s metric is Δ SPIDER-FL per the preregistration; a project-guide reference to “ Δ MACE”

was superseded (and its MACE description was inaccurate). (3) H3/H4 were preregistered on an AudioCaps-single-event set that was never acquired; we substitute paired per-clip comparisons on Clotho-eval, which are statistically stronger. (4) A planned “CLEAN” subset was never operationally defined; H1 uses the full 1,045-clip set, the same set anchoring the 0.261 threshold. (5) The polyphony detector is PANNs SED rather than the announced PaSST, because the definition requires frame-level output; the threshold fallback $\tau=0.50 \rightarrow 0.25$ followed the pre-committed rule. (6) H2’s “paired” wording is undefined for disjoint subsets; we use a two-sample bootstrap. (7) The preregistration also listed three descriptive-only probes (temporal ordering, out-of-distribution clips, a silence negative control); they were cut for scope and never run. No inferential hypothesis depends on them.

VI. CONCLUSION

Whether a zero-shot LALM beats trained audio captioners depends on the model: a current audio-specialist (AF3) does, significantly, on every overlap metric under our harness and above EnCLAP’s published SPIDER-FL, while a general omni-model and an older audio LALM do not. The advantage does not extend to grounding: AF3 hallucinates no less than SALMONN per caption (although it is the most grounded per mention), and caption-overlap gains on polyphonic clips are a dataset effect shared by all systems. Zero-shot captioning quality and audio grounding are separate axes, and future evaluation should report both. Three follow-ups seem most useful: temporally strong labels on real recordings, so that event ordering and counting can be scored next to captions; joint protocols that pair event-level and caption-level measurements on the same clips, extending the three-level idea beyond synthetic mixtures; and grounded metrics reported alongside overlap scores by default.

AI TRANSPARENCY STATEMENT

Per course requirements: the coding assistant Claude Code (Anthropic; Claude Opus 4.x and Claude Fable 5 models) was used throughout, under continuous human direction and review, for (i) pipeline, scoring, and analysis code, (ii) data analysis and visualization scripts, (iii) literature triage with primary-source verification, and (iv) LaTeX drafting and editing of this paper. Prompts ranged from implementation instructions with explicit verification requirements to review and audit requests; the repository’s session-dated logbook documents the work and the decisions taken under this assistance. All reported numbers are produced by committed scripts operating on raw model outputs; citations, APIs, and checkpoints were verified against primary sources (one published anchor was corrected during a final audit); all analysis decisions (thresholds, fallback rules, deviations) were fixed before results were inspected where noted, and are documented in the repository’s logbook and preregistration files.

CODE AND DATA AVAILABILITY

Code, configs, run manifests, result files, and the project logbook: <https://github.com/Zuraiz270/>

audio-to-text-captioning-using-LALMs. Clotho v2.1 is publicly available; model checkpoints are cited in the repository.

REFERENCES

- [1] K. Drossos, S. Lipping, and T. Virtanen, “Clotho: An audio captioning dataset,” in *Proc. IEEE ICASSP*, 2020, pp. 736–740.
- [2] A. Rohrbach, L. A. Hendricks, K. Burns, T. Darrell, and K. Saenko, “Object hallucination in image captioning,” in *Proc. EMNLP*, 2018, pp. 4035–4045.
- [3] S. Liu, Z. Zhu, N. Ye, S. Guadarrama, and K. Murphy, “Improved image captioning via policy gradient optimization of SPIDER,” in *Proc. IEEE ICCV*, 2017, pp. 873–881.
- [4] Z. Zhou, Z. Zhang, X. Xu, Z. Xie, M. Wu, and K. Q. Zhu, “Can audio captions be evaluated with image caption metrics?” in *Proc. IEEE ICASSP*, 2022, pp. 981–985.
- [5] D. Harish and J. Abeßer, “A three-level evaluation protocol for acoustic scene understanding of large language audio models,” in *Proc. DCASE Workshop*, 2025.
- [6] E. Labbé, “aac-metrics: Metrics for evaluating automated audio captioning systems,” <https://github.com/Labbeti/aac-metrics>, v0.5.5, 2024.
- [7] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, and M. D. Plumbley, “PANNs: Large-scale pretrained audio neural networks for audio pattern recognition,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 2880–2894, 2020.
- [8] F. Gontier *et al.*, “DCASE 2023 challenge task 6a: Automated audio captioning baseline system,” <https://github.com/felixgontier/dcase-2023-baseline>, 2023.
- [9] J. Kim, J. Jung, J. Lee, and S. H. Woo, “EnCLAP: Combining neural audio codec and audio-text joint embedding for automated audio captioning,” in *Proc. IEEE ICASSP*, 2024.
- [10] J. Kim, M. Jeon, J. Jung, S. H. Woo, and J. Lee, “EnCLAP++: Analyzing the EnCLAP framework for optimizing automated audio captioning performance,” in *Proc. DCASE Workshop*, 2024.
- [11] Y. Gong, Y.-A. Chung, and J. Glass, “AST: Audio spectrogram transformer,” in *Proc. Interspeech*, 2021, pp. 571–575.
- [12] J. F. Gemmeke, D. P. W. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, “Audio set: An ontology and human-labeled dataset for audio events,” in *Proc. IEEE ICASSP*, 2017, pp. 776–780.
- [13] J. Xu *et al.*, “Qwen2.5-Omni technical report,” arXiv:2503.20215, 2025.
- [14] C. Tang, W. Yu, G. Sun, X. Chen, T. Tan, W. Li, L. Lu, Z. Ma, and C. Zhang, “SALMONN: Towards generic hearing abilities for large language models,” in *Proc. ICLR*, 2024.
- [15] A. Goel, S. Ghosh, J. Kim, S. Kumar, Z. Kong, S.-g. Lee, C.-H. H. Yang, R. Duraiswami, D. Manocha, R. Valle, and B. Catanzaro, “Audio flamingo 3: Advancing audio intelligence with fully open large audio language models,” arXiv:2507.08128 (NeurIPS 2025), 2025.
- [16] J. Abeßer, S. Grollmisch, and M. Müller, “How robust are audio embeddings for polyphonic sound event tagging?” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 2658–2667, 2023.
- [17] S. Dixit, S. Deshmukh, and B. Raj, “MACE: Leveraging audio for evaluating audio captioning systems,” arXiv:2411.00321, 2024.
- [18] B. Efron, “Better bootstrap confidence intervals,” *Journal of the American Statistical Association*, vol. 82, no. 397, pp. 171–185, 1987.
- [19] N. L. Kerr, “HARKing: Hypothesizing after the results are known,” *Personality and Social Psychology Review*, vol. 2, no. 3, pp. 196–217, 1998.